

Sketching When an AI Tutor Should Abstain: A Study Protocol for Concept-Based Design of Abstention in Retrieval-Augmented Question Answering

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Abstract

Background. In earlier work, I found that a retrieval-augmented question-answering (RAG) system answered all 258 unanswerable questions in a 500-question SQuAD 2.0 sample, in the same form as its correct answers. A tutor that never says "I don't know" leaves learners no cue for calibrating their trust. Abstention is usually treated as a scoring problem: a learned confidence value decides when a model should decline. That framing hides the reasons a question should not be answered, which are also the reasons a learner would need to hear.

Aim. This protocol describes a study that uses model sketching, a method for composing machine learning models from human-understandable concepts, to design the decision of when an AI tutor should abstain.

Methods. Reusing the saved outputs of my earlier system, I will author six to ten concepts (for example, "the retrieved passage does not mention the question's main entity"), combine them into three to five sketch models, and compare these with a single retrieval-score threshold on the same 500 questions, then on a fresh held-out sample of 200. A dated design journal records how sketching changes my framing of the problem.

Planned analysis. Per-concept precision and recall, abstention F1 with bootstrap intervals for each sketch and the baseline, and a coded account of the design journal.

Status. Protocol only. No results have been collected. The study involves no human participants.

Keywords: abstention; model sketching; retrieval-augmented generation; AI tutors; trust calibration; human-centered machine learning

1. Introduction

Students increasingly ask AI systems questions about their course material. Whether that helps them depends less on how often the system is right than on whether they can tell when it might be wrong. One of the plainest cues a system can give is to decline: to say that it does not have a basis for an answer.

In my earlier project, a RAG pipeline built from standard components (a dense passage retriever, a FAISS index and Flan-T5-base) improved exact match from 0.004 to 0.062 over a closed-book baseline. It also answered every one of the 258 unanswerable questions in the sample. Its wrong answers to those questions looked the same as its correct answers to the others. For a learner, the second finding matters more than the first.

The usual fix is a calibrator: a model that scores how likely an answer is to be right and abstains below a threshold. A score can say that the system should decline, but not why. This study asks whether designing abstention from named reasons, rather than from a single score, shows problems that the score hides. I use model sketching (Lam et al., 2023), which lets a designer compose quick, functional models from concepts people can read and argue about.

2. Related work

Three lines of work meet in this study. Rajpurkar et al. (2018) introduced SQuAD 2.0, which adds questions with no answer in the passage, so that systems must learn when not to answer. Kamath et al. (2020) framed the decision as selective question answering and trained a calibrator to decide when a model should abstain, especially when test questions differ from training questions. Their decision is a learned score. My retriever came from Karpukhin et al. (2020); their dense passage retrieval finds relevant passages well, but retrieving a passage is not the same as having a basis to answer.

Model sketching (Lam et al., 2023) asks practitioners to compose high-level concepts into functional approximations of a model's logic, using zero-shot concept scoring. In a study with 17 practitioners, it shifted attention from implementation details to higher-level questions and helped surface gaps in problem formulation. It has been studied on tasks such as content moderation, not on abstention or education.

Table 1. Prior work and the gap this study addresses.

Line of work	What it offers	What it leaves open
Unanswerable questions (Rajpurkar et al., 2018)	A benchmark with questions that should not be answered	A benchmark, not a design method
Selective QA (Kamath et al., 2020)	A learned calibrator that decides when to abstain	The decision is a score, not reasons people can inspect
Dense retrieval (Karpukhin et al., 2020)	The retriever my pipeline used	Retrieval success is not the same as a basis to answer
Model sketching (Lam et al., 2023)	Concept-based, human-readable model design	Not yet applied to abstention or to education

3. Research questions

RQ1. Which human-understandable concepts separate the questions my tutor should have declined from the ones it should have answered?

RQ2. How do concept-based sketch models of abstention compare with a single retrieval-score threshold on the same questions?

RQ3. What does the sketching process reveal about the abstention problem that the original metrics (exact match, F1, Recall@K) did not?

4. Methods

4.1 Data. The study reuses the evaluation run from my earlier project; no new data is collected. The run used a fixed, seeded sample of 500 questions from the SQuAD 2.0 validation split (242 answerable, 258 unanswerable). The retriever was Dense Passage Retrieval with its Natural Questions checkpoints, over an exact inner-product FAISS index of about 29,000 chunks drawn from about 20,000 SQuAD 2.0 passages. The generator was Flan-T5-base with beam search, and its prompt contained no instruction about unanswerable questions. For each question I need the question, the top-K passages and their scores, the generated answer, and the gold label. If only summary metrics were saved, I will repeat the run with the same seed.

4.2 Baseline. I will fit a single threshold on the top-1 retrieval score to predict whether the system should abstain, using the unanswerable label as ground truth.

4.3 Concept authoring. Using the open-source ModelSketchBook implementation, I will write six to ten concepts as short natural-language descriptions. Starting candidates are:

- The retrieved passage does not mention the question's main entity.
- The question asks about something the passage negates or contradicts.
- The question contains a false premise.
- The top retrieved passages disagree with each other.
- The answer would require information outside the passage.

4.4 Sketching. I will combine concepts into three to five sketch models: single concepts, small sets, and all concepts together. After each sketch I will read 20 misclassified questions and

revise, drop, or add concepts. Every change is logged in a dated design journal with the reason for it.

4.5 Held-out check. To guard against tuning concepts to the 500 questions, I will draw a fresh seeded sample of 200 SQuAD 2.0 validation questions, run the original pipeline on them, and evaluate only the final sketch and the baseline.

5. Planned analysis

RQ1 will be answered with per-concept precision and recall against the abstain label, plus a table of example questions each concept catches and misses. RQ2 will be answered with abstention accuracy, precision, recall and F1 for each sketch and the baseline, with bootstrap 95% intervals over the 500 questions and a single comparison on the held-out 200. RQ3 will be answered from the design journal: I will code entries for moments when sketching changed how I framed the problem (a new concept, a dropped concept, or a question I relabeled after reading it).

6. Limitations

- There is one designer (me), so RQ3 is a reflective design account, not a user study.
- SQuAD questions are not student questions, and SQuAD passages are not course materials.
- Zero-shot concept scores depend on the underlying model, and results may not transfer to another one.
- No learners are involved. Any claim about learner trust is an implication of the findings, not a finding.

7. Status and timeline

As of October 2026, this is a protocol; no part of the study has been run. I plan to check the saved outputs in mid-October, author concepts and the baseline by early November, sketch

through mid-November, run the held-out check by late November, and post a short technical report by November 30, 2026. The concepts, sketch models and analysis code will be shared with the report.

Declarations

Use of AI tools. Claude (Anthropic) assisted with drafting the structure and wording of this protocol and with checking references against Crossref records. I chose the study, made the design decisions, and reviewed and revised the text. I take full responsibility for its content.

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